Property Assessment

Entities

Property, Assessment, Sale, Appeal

Attributes

Property: property ID, age (year built), address, tax code area, zone code, subdivision, construction price, township/range/section, GPS latitude, GPS longitude, GPS altitude

Estimate: valuation, method, lot size, house size, neighborhood, general condition, pictures from the road, remodel year, sewer, water, utilities, bedrooms, bathrooms, fireplaces, garage, basement, comments

Sale: permanent sale price, year

Appeal: date, claimed value, resulting value comments

Relationships

Assumption: A Property can have many Estimates. An Estimate is for a single Property.

- Cardinality: Property 1:M Assessment
- Multiplicity:
    - A Property may be Assessed (optional: min 0)
    - A Property can be Assessed many times (max: many)
    - An Assessment must be for a Property (mandatory: min 1)
    - An Assessment is only for a single Property (max: 1)

Assumption: A Property can have many Inspections. An Inspection is for a single Property.

- Cardinality: Property 1:M Assessment
- Multiplicity:
    - A Property may be Assessed (optional: min 0)
    - A Property can be Assessed many times (max: many)
    - An Assessment must be for a Property (mandatory: min 1)
    - An Assessment is only for a single Property (max: 1)

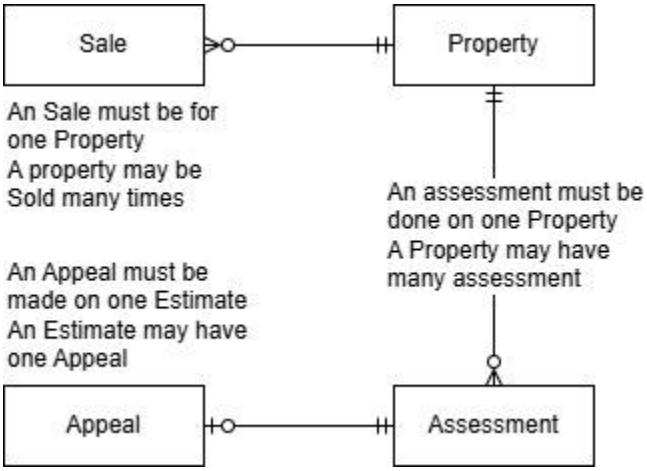
Assumption: A Property can have many Sales. A Sale is for a single Property.

- Cardinality: Property 1:M Sale
- Multiplicity:
    - A Property may be Sold (optional: min 0)

- A Property can be Sold many times (max: many)
- A Sale must be for a Property (mandatory: min 1)
- A Sale is only for a single Property (max: 1)

Assumption: A Property can have many Appeals. An Appeal is for a single Property.

- Cardinality: Property 1:M Appeal
- Multiplicity:
  - A Property may be Appealed (optional: min 0)
  - A Property can be Appealed many times (max: many)
  - An Appeal must be for a Property (mandatory: min 1)
  - An Appeal is only for a single Property (max: 1)



Music Store

Entities

Customer, Buyer Level, Sale, Inventory

Attributes

Customer: customer ID, first name, last name, band, email address, phone, reward level, delivery address – street address, city, state, zip code

Sale: sale ID, date, employee, item, quantity, discount/eligibility (are sales themselves ineligible or just the items?), total price

Item: name, description, price, discount eligibility

Buyer Level: level, yearly sales total, discount, name recognition (get clarification on what this is)

Relationships

Assumption: A Customer can have many Sales. A Sale is made by a single Customer.

- Cardinality: Customer 1:M Sale
- Multiplicity:
    - A Customer may have a Sale (optional: min 0)
    - A Customer can have many Sales (max: many)
    - A Sale must be for a Customer (mandatory: min 1)
    - A Sale is only for a single Customer (max: 1)

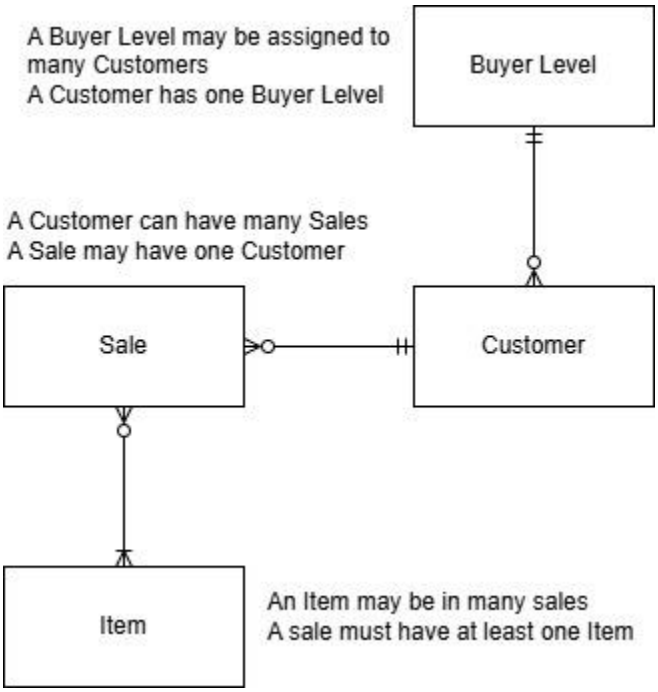
Assumption: A Buyer Level can be assigned to many Customers. A Customer is assigned one Buyer Level.

- Cardinality: Buyer Level 1:M Customer
- Multiplicity:
    - A Buyer Level may be associated to a Customer (optional: min 0)
    - A Buyer Level can be associated to many Customers (max: many)
    - A Customer must have a Buyer Level (mandatory: min 1)
    - A Customer only has one Buyer Level (max: 1)

Assumption: An Item can be in many Sales. A Sale has many Items.

- Cardinality: Items M:N Sale
- Multiplicity:
    - An Item may be Sold (optional: min 0)

- An Item can be Sold many times (max: many)
- A Sale must include at least one item (mandatory: min 1)
- A Sale may include many Items (max: 1)



Document Translation

Entities

Translator, Client, Contract, Document, Language

Attributes

Translator: name, languages, native language, skill levels, phone, email (is tracking the native language necessary?)

Client: name, phone, address – street address, city, state, zip code, email, department (does this belong to the client?)

Contract: client, translator, date, target language, documents, % commission

Document: name, date received, language, date due, intial date, review date, price, difficulty

Language: name

Relationships

Assumption: A Translator must speak multiple Languages. A Language may be spoken by many Translators.

- Cardinality: Translator 1:M Language
- Multiplicity:
    - A Translator must speak at least two Languages (mandatory: min 2)
    - A Translator can speak many Languages (max: many)
    - A Language may be spoken by a Translator (optional: min 0)
    - A Language may be spoken by many Translators (max: many)

Assumption: A Translator can have many Contracts. A Contract is given to one Translator.

- Cardinality: Translator 1:M Contract
- Multiplicity:
    - A Translator may have Contracts (optional: min 0)
    - A Translator can be associated to many Contracts (max: many)
    - A Contract must have a Translator (mandatory: min 1)
    - A Contract can only have one Translator (max: 1)

Assumption: A Client can have many Contracts. A Contract has one Client.

- Cardinality: Client 1:M Contract

- Multiplicity:
    - A Client may have Contracts (optional: min 0)
    - A Client can have many Contracts (max: many)
    - A Contract must have a Client (mandatory: min 1)
    - A Contract has only one Client (max: 1)

Assumption: A Contract is for many Documents. A Document must be in one Contracts.

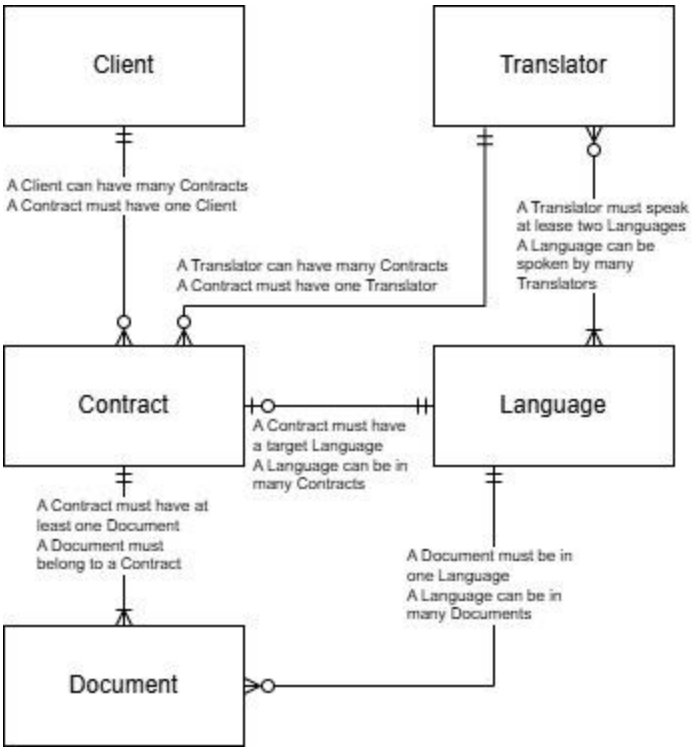
- Cardinality: Contract 1:M Document
- Multiplicity:
    - A Contract must have a Document (mandatory: min 1)
    - A Contract can have many Documents (max: many)
    - A Document must be in a Contract (mandatory: min 1)
    - A Document can be in no more than one Contracts (max: 1)

Assumption: A Language can be for many Contracts. A Contract is for one Language.

- Cardinality: Language 1:M Contract
- Multiplicity:
    - A Language may be in a Contract (optional: min 0)
    - A Language can be in many Contracts (max: many)
    - A Contract must have a Language (mandatory: min 1)
    - A Contract has only one Language (max: 1)

Assumption: A Language can be in many Documents. A Document is in one Languages.

- Cardinality: Language M:N Contract
- Multiplicity:
    - A Language can be used in a Document (optional: min 0)
    - A Language can be used in many Document (max: many)
    - A Document must be in a Language (optional: min 1)
    - A Document can only be in one Language (max: 1)


Client
Translator
A Client can have many Contracts
A Contract must have one Client
A Translator can have many Contracts
A Contract must have one Translator
A Translator must speak
at lease two Languages
A Language can be
spoken by many
Translators
Contract
Language
A Contract must have
a target Language
A Language can be in
many Contracts
A Contract must have at
least one Document
A Document must
belong to a Contract
A Document must be in
one Language
A Language can be in
many Documents
Document